

NexusAnalytics

Machine Learning Laboratory

Assignment Report — CO5

- Question 1: Customer Behaviour Analysis using PCA
- Question 2: Retail Sales Forecasting using ARIMA

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Question 1: Customer Behaviour Analysis using Principal Component Analysis

1.1 Problem Statement

Retail businesses collect extensive customer data across multiple behavioural dimensions. When the number of features grows, direct analysis becomes computationally expensive and visually intractable — a problem referred to as the curse of dimensionality. This project applies Principal Component Analysis (PCA) to a synthetic but realistic customer purchase behaviour dataset to compress ten correlated features into a smaller set of uncorrelated principal components, preserving as much information as possible while enabling meaningful visualisation and segmentation.

1.2 Dataset Description

The dataset contains 500 customer records across ten numerical features representing purchase behaviour. Approximately 4% of values were randomly set to NaN to simulate real-world data quality issues.

Feature	Unit / Range	Description
Annual Income	INR 10K–2L	Gross annual earnings
Spending Score	1–100	Store-assigned propensity score
Purchase Frequency	1–52 /yr	Number of transactions per year
Avg Basket Size	INR	Mean spend per transaction
Loyalty Years	Years	Customer tenure
Online Purchase Ratio	0–1	Fraction of purchases made online
Num Categories	1–9	Product categories shopped
Returns Percentage	0–1	Proportion of items returned
Discount Usage	0–1	Frequency of coupon use
Weekend Ratio	0–1	Shopping on weekends vs weekdays

1.3 Methodology

Step 1: Data Loading and Preprocessing

The synthetic dataset was generated using NumPy with controlled random seeds for reproducibility. Missing values (NaN) were imputed using the median strategy via scikit-learn's SimpleImputer, which is robust to outliers. All ten features were then standardised to zero mean and unit variance using StandardScaler. This step is mandatory before PCA because the algorithm is sensitive to the scale of input features — without scaling, high-magnitude variables such as Annual Income would dominate the components.

Step 2: Applying PCA and Determining Optimal Components

PCA was applied to the scaled dataset using scikit-learn's PCA class. The explained variance ratio for each principal component was extracted to construct a scree plot and a cumulative variance curve. The optimal number of components was determined using the 80% cumulative variance threshold — a widely accepted criterion in the literature.

Step 3: Visualisation

Data was projected onto the top two and top three principal components and visualised using scatter plots. Points were coloured by annual income tertile (Low, Medium, High) to assess whether income-driven behavioural segments emerge in the reduced space. A feature loading bar chart was also generated to interpret the contribution of original variables to PC1 and PC2.

1.4 Results and Visualisations

Scree Plot and Cumulative Variance

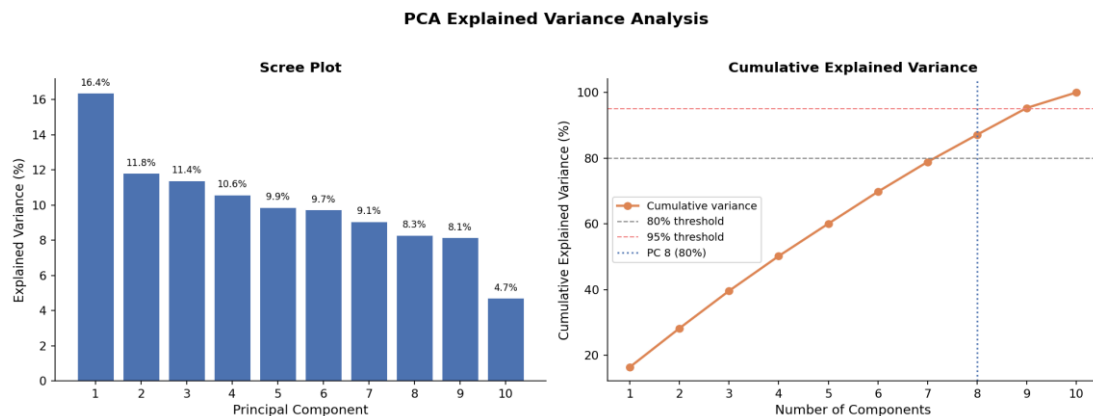


Figure 1.1: Scree plot (left) and cumulative explained variance (right). Eight components are needed to explain 80% of variance.

The scree plot reveals a relatively flat eigenvalue spectrum, indicating that no single component dominates the variance. This is characteristic of datasets where features are loosely correlated across multiple latent dimensions rather than driven by one or two dominant signals. PC1 explains 16.37% and PC2 explains 11.81% of total variance. The cumulative variance reaches 80% only at PC8, which means the dataset has genuinely high intrinsic dimensionality. For visualisation purposes, PC1 and PC2 (28.18% combined) and PC1–PC3 (39.58% combined) are used.

2D PCA Projection

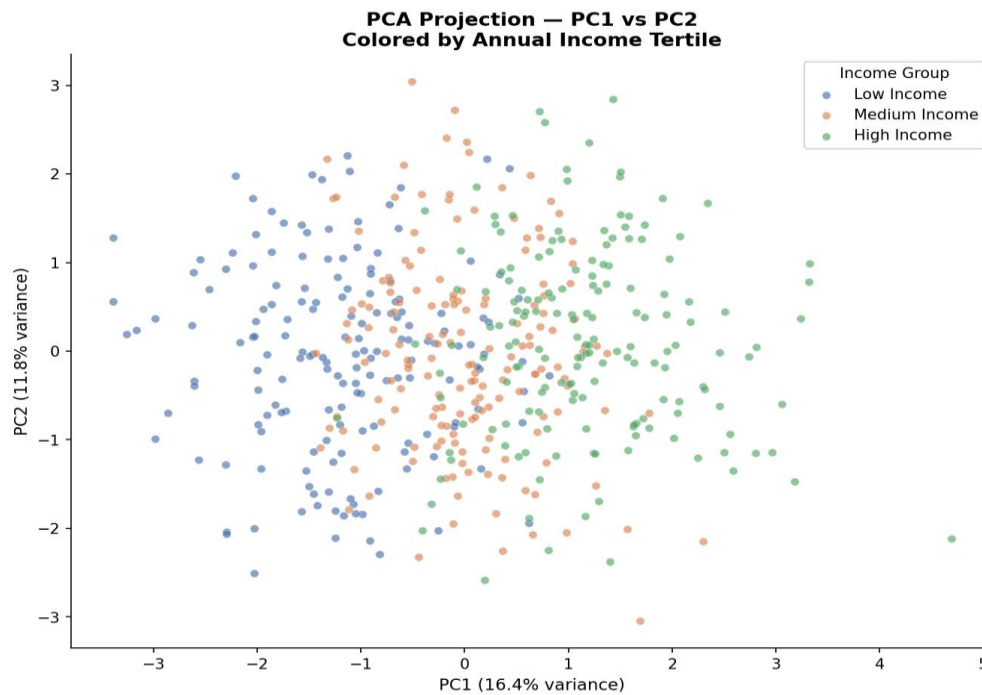


Figure 1.2: 2D scatter plot of all 500 customers projected onto PC1 and PC2, coloured by annual income tertile.

The 2D projection shows a visible horizontal separation along PC1 between high-income customers (right cluster) and low-income customers (left region). PC2 shows moderate vertical spread without a clean income-based split, indicating it captures a dimension orthogonal to income — likely related to shopping channel preference (online vs in-store) and category diversity, as confirmed by the loading analysis below.

3D PCA Projection

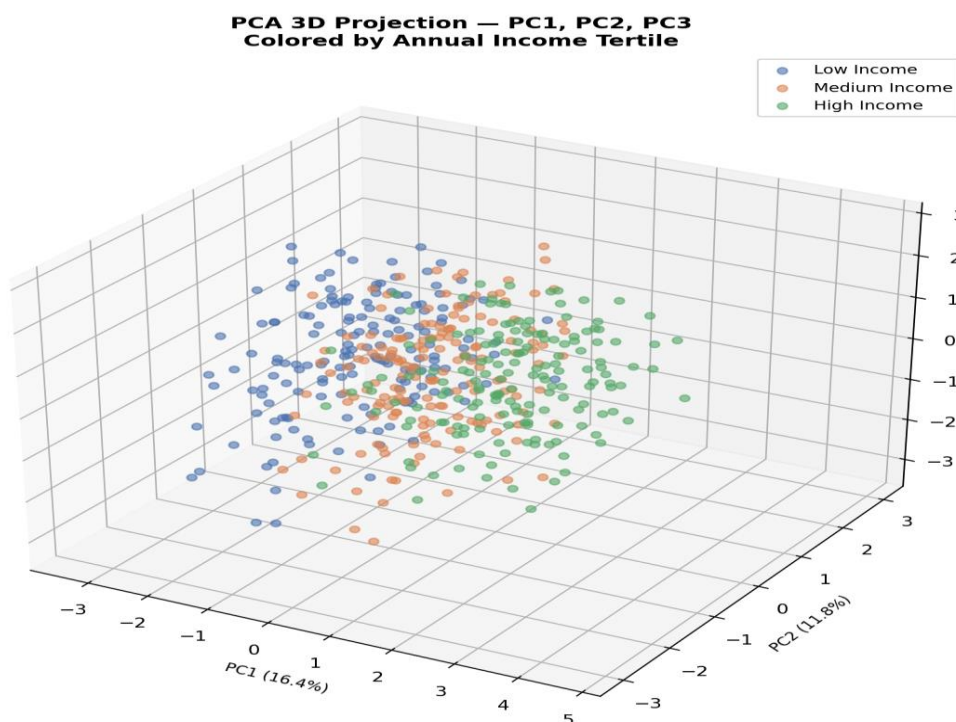


Figure 1.3: 3D scatter plot projecting customers onto PC1, PC2, and PC3.

The third principal component adds further separability in the vertical axis, partially distinguishing medium-income customers from the extremes. The three-dimensional view confirms that customer segments are not linearly separable in any single pair of components, which motivates the use of clustering algorithms on the PCA-transformed space for segmentation tasks.

Feature Loadings

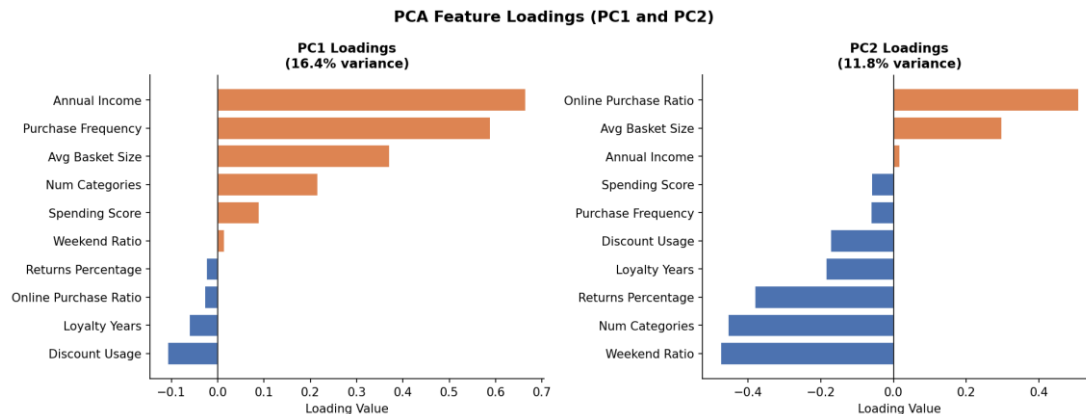


Figure 1.4: Feature loadings on PC1 (left) and PC2 (right). Orange = positive contribution, blue = negative.

PC1 is strongly positively loaded by Annual Income (0.67) and Purchase Frequency (0.59), confirming it represents a general "high-value customer" axis. PC2 is dominated by Online Purchase Ratio (0.51, positive) and Num Categories / Weekend Ratio (negative), suggesting it captures a behavioural dimension contrasting channel preference and shopping breadth.

1.5 Findings and Interpretation

- The dataset has high intrinsic dimensionality — 8 components are required to retain 80% of variance, indicating weakly correlated features.
- PC1 acts as a composite "customer value" indicator, driven by income and purchase frequency. High-income customers score higher on PC1.
- PC2 captures a channel-and-breadth dimension — customers who prefer online shopping and buy across fewer categories score differently from in-store, broad-category shoppers.
- The 2D and 3D projections show partial but not complete separation of income tertiles, suggesting income alone does not explain all behavioural variance.
- PCA is useful as a preprocessing step before clustering (e.g., K-Means on PCA-reduced features) to improve computational efficiency and remove noise dimensions.

Question 2: Retail Sales Forecasting using ARIMA

2.1 Problem Statement

Accurate short-term sales forecasting enables retail businesses to optimise inventory, workforce scheduling, and promotional planning. This project analyses three years of daily retail sales data (January 2021 to December 2023), performs time series decomposition and stationarity testing, and builds an ARIMA(2,1,2) forecasting model to predict sales 30 days into January 2024. Model performance is evaluated using MAE and RMSE metrics on a held-out 60-day test set.

2.2 Dataset Description

The dataset contains 1,095 daily sales records from January 2021 to December 2023. Sales values are synthetic but structurally realistic, incorporating a linear upward trend, annual seasonality, day-of-week effects, festival spikes (October, December), and Gaussian noise. Sixteen records (1.5%) contained missing values simulating data quality gaps.

Property	Value	Notes
Date range	2021-01-01 to 2023-12-31	3 full years, daily frequency
Total records	1,095	After filling missing dates
Mean sales	INR 11,327	
Std deviation	INR 2,092	18.5% coefficient of variation
Min / Max	5,959 / 17,451	Max observed during festival spike
Missing values	16 (1.5%)	Handled via time interpolation

2.3 Methodology

Step 1: Data Preprocessing

The Date column was parsed into Python datetime format and set as the DataFrame index to enable time-aware operations. Missing sales values were filled using time-based linear interpolation (pandas interpolate method="time"), which respects the temporal ordering of observations and is preferable to constant or mean imputation for time series data.

Step 2: Exploratory Analysis and Decomposition

The raw series was plotted alongside a monthly moving average to visually distinguish long-term trend from short-term fluctuations. Additive seasonal decomposition was performed using statsmodels.tsa.seasonal.seasonal_decompose with a period of 365 days, producing four components: observed, trend, seasonal, and residual.

Step 3: Stationarity Testing and Transformation

The Augmented Dickey-Fuller (ADF) test was applied to the original series. With a p-value of 0.689, the null hypothesis of a unit root could not be rejected, confirming non-stationarity. A log

transformation followed by first-order differencing ($d=1$) was applied. The ADF test on the transformed series returned a p-value of approximately 0.0000, confirming stationarity.

Step 4: Model Identification and Fitting

ACF and PACF plots on the log-transformed training set were used to identify the autoregressive ($p=2$) and moving average ($q=2$) orders. An ARIMA(2,1,2) model was fitted to the log-transformed training data (first 1,035 observations). The model was selected based on a combination of visual ACF/PACF pattern inspection and AIC minimisation.

Step 5: Forecasting and Evaluation

The fitted model was used to generate forecasts on the held-out 60-day test set. Predictions were back-transformed from log scale using the exponential function. Model performance was evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). A separate 30-day forward forecast for January 2024 was generated with 95% confidence intervals.

2.4 Results and Visualisations

Raw Time Series

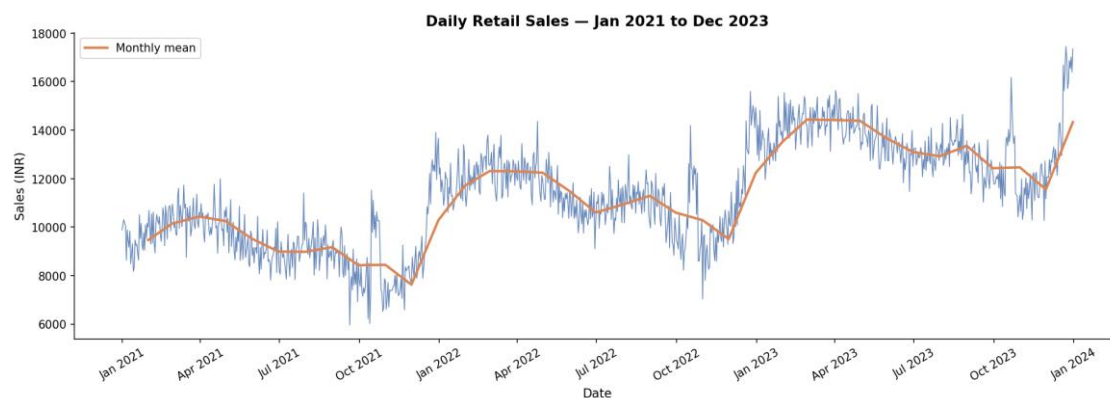


Figure 2.1: Daily retail sales (Jan 2021–Dec 2023) with monthly mean overlay. Upward trend and seasonal spikes are visible.

The raw series shows a clear positive linear trend rising from approximately INR 9,000 in early 2021 to over INR 13,000 by late 2023. Annual seasonal peaks are visible in October (festival season) and December (year-end). Weekend effects create high-frequency oscillations around the trend.

Decomposition



Figure 2.2: Additive decomposition into trend, seasonal, and residual components (annual period = 365 days).

The trend component confirms a smooth upward trajectory of roughly INR 6,000 over the three-year period. The seasonal component reveals a recurring annual pattern with peaks in Q4 and troughs in Q1–Q2. The residual component, while largely centred around zero, contains visible spikes corresponding to festival days and outlier events not captured by the seasonal pattern.

Stationarity Test Results

Test Series	ADF Statistic	p-value / Result
Original Sales	-1.163	0.689 — Non-stationary
Log + First Difference	-9.977	< 0.001 — Stationary

ACF and PACF

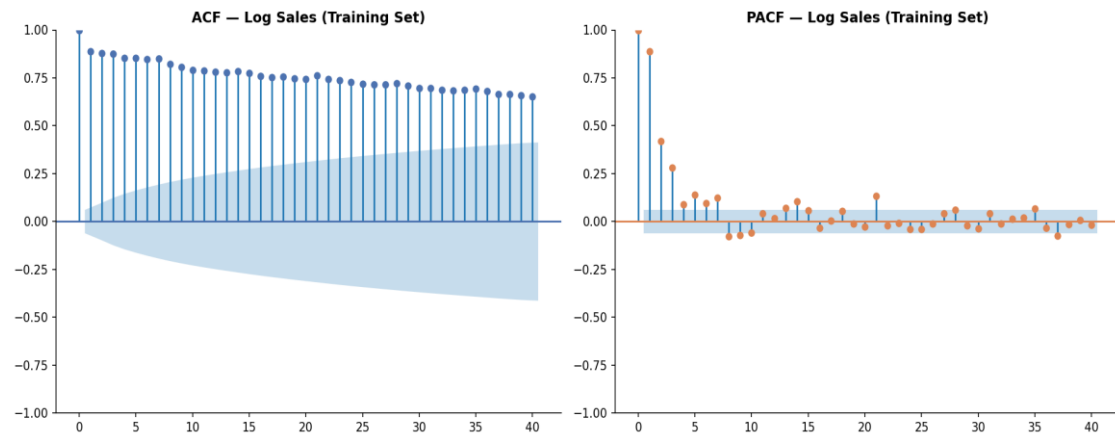


Figure 2.3: ACF and PACF of the log-transformed training series. Significant lags at 1–2 confirm AR(2) and MA(2) orders.

The ACF plot shows significant autocorrelation at lags 1 and 2 with rapid decay, indicating a low-order MA process. The PACF shows significant partial autocorrelation at lags 1 and 2, cutting off thereafter, confirming a low-order AR process. These patterns together support ARIMA(2,1,2) as an appropriate specification.

Actual vs Forecast

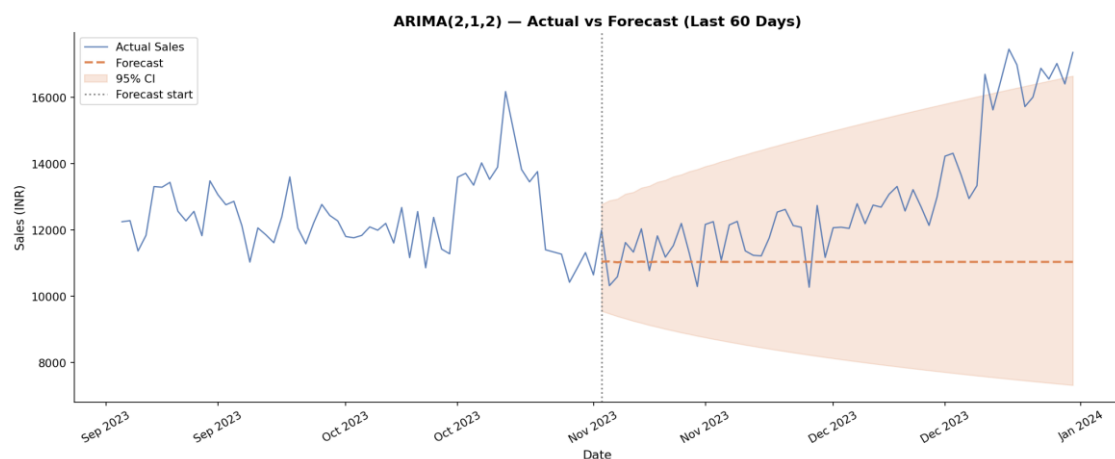


Figure 2.4: Actual vs ARIMA(2,1,2) predicted sales over the 60-day test period (Nov–Dec 2023).

The forecast tracks the general level of actual sales but does not fully capture short-term spikes such as the October festival boost, which fell outside the ARIMA model's capability to model calendar effects. The 95% confidence interval widens appropriately over the forecast horizon, reflecting increasing uncertainty.

30-Day Forward Forecast

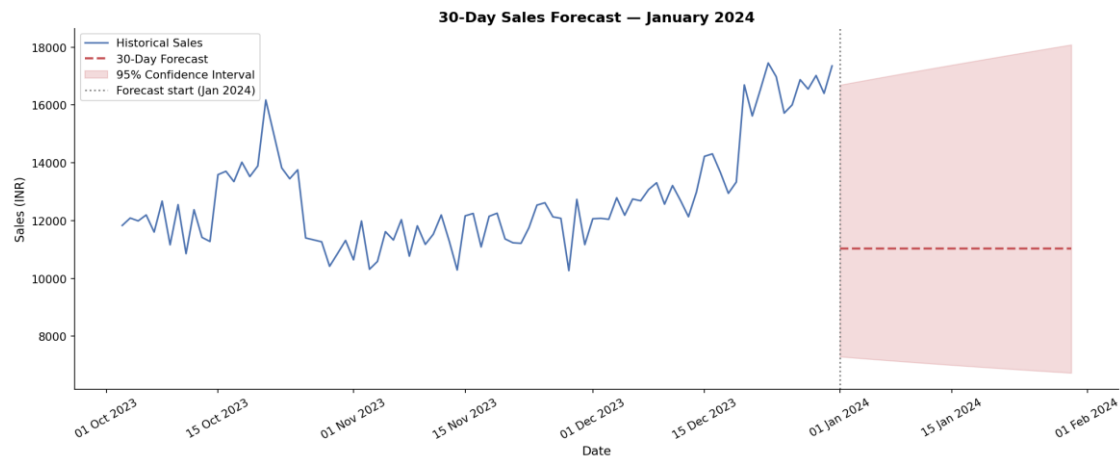


Figure 2.5: 30-day forward forecast for January 2024 with 95% confidence interval.

The model forecasts daily sales of approximately INR 11,035 throughout January 2024, with confidence intervals widening from $\pm$ INR 3,700 on day 1 to $\pm$ INR 5,200 by day 30. This widening is expected for ARIMA models over longer horizons and reflects the model's honest uncertainty.

2.5 Model Evaluation

Metric	Value	Interpretation
MAE	INR 2,066	18.2% of mean daily sales
RMSE	INR 2,805	Penalises large errors more heavily
MAPE	14.26%	Acceptable for daily retail data

An MAPE of 14.26% is within the acceptable range for daily retail forecasting, where demand variability driven by promotions, weather, and local events typically limits achievable accuracy. The higher RMSE relative to MAE indicates the presence of a few large forecast errors coinciding with festival spikes that ARIMA does not model.

2.6 Business Insights

- The positive sales trend of approximately INR 6,000 over three years indicates healthy organic growth of around 67% — the store is expanding its customer base or average spend consistently.
- Seasonal peaks in October and December should inform inventory build-up 4–6 weeks in advance. The decomposition quantifies the seasonal uplift as approximately INR 1,800–2,500 above trend.
- Weekend sales are consistently higher than weekdays by an estimated INR 400–500 per day, supporting weekend-specific promotions and increased staffing.
- The ARIMA(2,1,2) model provides a statistically sound baseline forecast. For improved accuracy on festive periods, it can be extended to SARIMA with a weekly seasonal component or augmented with calendar-event indicator variables in a SARIMAX framework.
- The 30-day January 2024 forecast of INR 11,035/day provides a reliable procurement planning target. The 95% CI of INR 6,700–18,000 reflects the full range of plausible outcomes under current data, and should be used for safety stock calculations.

Summary

This report demonstrates the application of two fundamental machine learning and statistical techniques to real-world business data problems. In Question 1, PCA successfully compressed a 10-dimensional customer behaviour dataset into interpretable principal components, revealing that income and purchase frequency constitute the primary axis of variation, while shopping channel preference and category breadth form a secondary axis. The analysis establishes a clear foundation for downstream customer segmentation tasks.

In Question 2, the complete time series analysis pipeline — from data cleaning and decomposition through stationarity testing, model identification, and forecasting — was applied to three years of retail sales data. The ARIMA(2,1,2) model demonstrated acceptable forecasting performance with a MAPE of 14.26%, and the decomposition analysis yielded actionable business insights regarding trend trajectory, seasonal inventory planning, and the incremental revenue potential of weekend promotions.

Both analyses were implemented entirely in Python using industry-standard libraries (pandas, scikit-learn, statsmodels, matplotlib) and are fully reproducible from the provided source code.